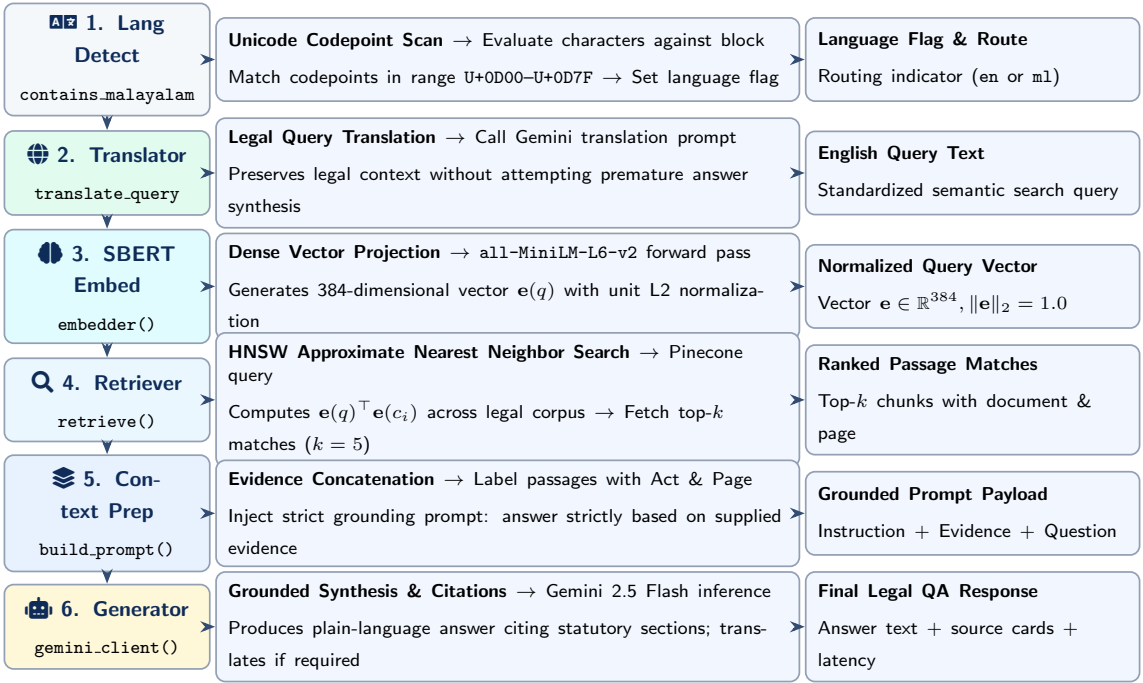

1. Lang Detect
contains_malayalam
Unicode Codepoint Scan → Evaluate characters against block
Match codepoints in range U+0D00–U+0D7F → Set language flag
Language Flag & Route
Routing indicator (en or ml)
2. Translator
translate_query
Legal Query Translation → Call Gemini translation prompt
Preserves legal context without attempting premature answer synthesis
English Query Text
Standardized semantic search query
3. SBERT Embed
embedder()
Dense Vector Projection → all-MiniLM-L6-v2 forward pass
Generates 384-dimensional vector $\mathbf{e}(q)$ with unit L2 normalization
Normalized Query Vector
Vector $\mathbf{e} \in \mathbb{R}^{384}$, $\|\mathbf{e}\|_2 = 1.0$
4. Retriever
retrieve()
HNSW Approximate Nearest Neighbor Search → Pinecone query
Computes $\mathbf{e}(q)^\top \mathbf{e}(c_i)$ across legal corpus → Fetch top-k matches ($k = 5$)
Ranked Passage Matches
Top-k chunks with document & page
5. Context Prep
build_prompt()
Evidence Concatenation → Label passages with Act & Page
Inject strict grounding prompt: answer strictly based on supplied evidence
Grounded Prompt Payload
Instruction + Evidence + Question
6. Generator
gemini_client()
Grounded Synthesis & Citations → Gemini 2.5 Flash inference
Produces plain-language answer citing statutory sections; translates if required
Final Legal QA Response
Answer text + source cards + latency